

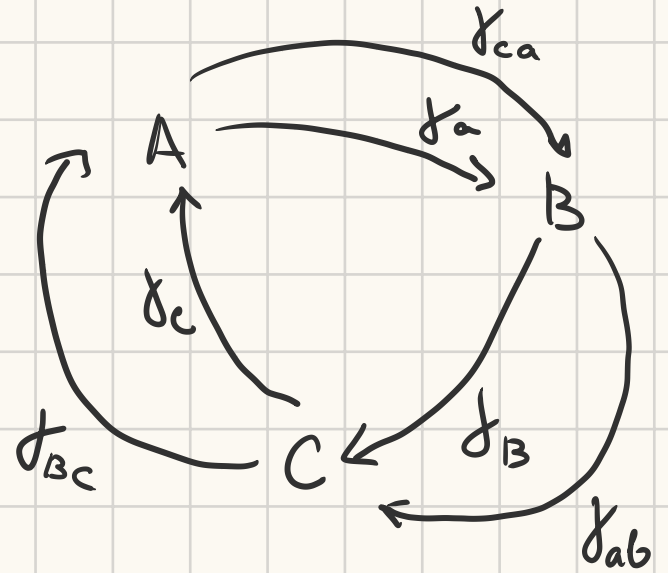
Обмен ключами: (D-H)

2-раундовый обмен:
ключами внутри группы

$$G = \langle g \rangle$$

1 круг: Внутр
2 круг: Внешн

$$\begin{aligned} A &\rightarrow \text{з.к } a \in \mathbb{Z} \\ B &\rightarrow \text{з.к } b \in \mathbb{Z} \\ C &\rightarrow \text{з.к } c \in \mathbb{Z} \end{aligned}$$



1-раундовый вариант:

$$G \times G \rightarrow H, \quad G, H - \text{цикл. группы.}$$

$$\langle g, g_2, g \rangle = \langle g, g \rangle \langle g_2, g \rangle$$

$$\langle g, g, g_2 \rangle = \langle g, g_1 \rangle \langle g, g_2 \rangle$$

$$A = \langle g^b, g^{ca} \rangle$$

$$B = \langle g^a, g^{cb} \rangle$$

$$C = \langle g^b, g^{ac} \rangle$$

$\exists \langle g, g \rangle$ - порождающая H

ID-сеть:

$$\text{ОКО: } G = \langle g \rangle; \langle \cdot, \cdot \rangle: G \times G \rightarrow H$$

секрет $s \in \mathbb{Z}$ | $H_1: \underbrace{\{0,1\}^*}_{\text{хеш функция}} \rightarrow G$ $H_2: H \rightarrow \{0,1\}^l$

Открыто $x = g^s$

$$\forall \text{ в сети имеют ID} \in \{0,1\}^*$$

$$A \rightarrow \text{IDA и вычисляет} \rightarrow K_A = H_1(\text{IDA})$$

$$\rightarrow C_A = H_1(\text{IDA})^s$$

B хочет отправить сообщ. M:

$$(C_1, C_2) = (g^\epsilon, M \oplus H_2(\langle K_A, K \rangle^\epsilon)), \epsilon \in \mathbb{Z}$$

Расшифровка?

$$C_2 \oplus H_2(\langle C_A, C_1 \rangle) = M \oplus H_2(\langle K_A, K \rangle^\epsilon) \oplus H_2(\langle C_A, C_1 \rangle) \oplus$$

$$\oplus M \oplus \underbrace{H_2(\langle H_1(\text{IDA}), g^s \rangle^\epsilon) \oplus H_2(\langle (H_1(\text{IDA}))^s, g^\epsilon \rangle)}_{\text{по свойству = 0 ввиду симметрии}} = \textcircled{M}$$

Дивизоры на кривых:

E - гладкая проективная кривая/ K

$$\text{Div}(E) \stackrel{\text{def}}{=} \left\{ \sum_{P \in E(K)}^{\text{конечная}} a_i [P_i] \mid a_i \in \mathbb{Z} \right\} - \text{группа по } \oplus$$

группа дивизоров (Вейля) $[P_i]$ - символ $\rightarrow P_i$

$$\text{Div}(E) \xrightarrow{\deg} \mathbb{Z}; \sum_i a_i [P_i] \mapsto \sum_i a_i$$

$$\ker(\deg) = \text{Div}^0(E)$$

$$\text{Пр: } \overline{\mathbb{R}} = \mathbb{C}; \overline{\mathbb{F}_p} = \bigcup_{i \geq 1} \mathbb{F}_{p^i}$$

$$\text{Div}^0(E) \xrightarrow{\Phi} E(\overline{K}); \overline{K} - \text{алг. замык. поля } K$$

$$\Phi: [P] - [\infty] \mapsto P; \quad \Phi - \text{сюръективный гомоморфизм групп}$$

$\uparrow$
беск. точка

$$\text{Div}^0(E) / \ker(\Phi) \cong E(\overline{K})$$

$\ker(\Phi)$ (Th Абеля)

Главные дивизоры

$f \in K(x, y)$ - рационал
ф-я

$f \in \overline{K}(E)$ - ф-я, если она опред. хоть в какой-то (!) кривой

$$\text{Пример: } E = \{y^2 = x^3 + Ax + B\}, \quad y^2 = x^3 - x, \quad \frac{x}{y} \in \overline{K}(E)$$

$$\frac{x}{y} = \frac{y}{x^2 - 1} \quad - \text{опред в } (0, 0)$$

$\frac{x}{y}$ не опред в $(0, 0)$

$$f \in \overline{K}(E), f \text{ имеет нуль в } P, \text{ если } f(P) = 0$$

$$P \in E \quad f \text{ имеет полюс в } P, \text{ если } "f(P) = \infty" \Leftrightarrow \frac{1}{f} \text{ имеет нуль в } P$$

$\text{ord}_P(f)$ - порядок $\rightarrow$ нуля или полюса f .

$$P \mapsto u_P - \text{ф-я (лок.-я параметр)} \quad u_P(P) = 0 \quad f = u_P^{\text{ord}_P(f)} g, \quad P - \text{не полюс гл. } g$$

$$\text{ord}_{(0,0)}(x) = 2 \quad x = y^2 \cdot \frac{1}{x^2 - 1} \quad y = u(0,0)$$

$$\text{ord}_{(0,0)}\left(\frac{x}{y}\right) = 1$$

$$2) E = \{y^2 = x^3 + Ax + B\} \quad u_\infty = \frac{x}{y}; \left(\frac{x}{y}\right)^2 = \frac{1}{x} \left(1 + \frac{A}{x^2} + \frac{B}{x^3}\right)^{-1} \quad \text{ord}_\infty(x) = -2$$

3) $\mathbb{P}^1$; $f = x_0 x_1$ $g = (x_0 - x_1)^2$
 $(x_0 : x_1)$
 $P_0 = (1 : 0)$ $\text{ord}_{P_0}(f) = 1$
 $P_1 = (0 : 1)$ $Q = (1 : 1)$ $\text{ord}_Q(g) = 2$

$$f \in \bar{K}(E) \rightarrow \text{div}(f) = \sum_{P \in \bar{K}(E)} \text{ord}_P(f) [P] \in \text{Div}(E)$$

$$P\text{Div}(E) = \left\{ \sum \alpha_i \text{div}(f_i) \mid \alpha_i \in \mathbb{Z}; f_i \in \bar{K}(E) \right\} - \text{подгруппа } \text{Div}^0(E)$$

Сб-ба: 1) $\deg(\text{div}(f)) = 0$ $f = \frac{R(x, y, z)}{S(x, y, z)}$: R, S - од-не мн-м степеней d
 Отыска: $\deg(\text{div}(f)) = d - d = 0$

2) $\text{div}(fg) = \text{div}(f) + \text{div}(g)$
 $f, g \in \bar{K}(E)$ $\left(\bar{K}(E) \xrightarrow[\text{гомоморфизм}]{\text{div}} \text{Div}(E) \right)$ $\text{div}\left(\frac{1}{f}\right) = -\text{div}(f)$

3) Если f без нулей и полюсов, то $f = \text{const}$
 (Th. Лувуна)

Th: Аделей: $\frac{\text{Div}^0(E)}{P\text{Div}(E)} \xrightarrow{\varphi} \bar{E}(K)$ эллиптическая кривая, кубика
 $\mathcal{C}^0(E)$ - группа классов дивизоров (Бетина)

Пример: 1) $\mathbb{P}^1$ $f = x_0 x_1$ $g = (x_0 - x_1)^2$ $\varphi = \frac{f}{g} \in K(\mathbb{P}^1)$ $\text{div}(\varphi) = \text{div}(f) - \text{div}(g) =$
 $= [P_0] - [P_1] - 2[Q]$

$\mathcal{C}^0(E) = \frac{\text{Div}(E)}{P\text{Div}(E)}$ $\text{Div}(\mathbb{P}^1) \xrightarrow{\deg} \mathbb{Z}$ deg-сюръекция
 $\mathcal{C}^0(P_1) \cong \mathbb{Z}$ $\ker(\deg) = P\text{Div}(\mathbb{P}^1)$
 $\exists \sum \alpha_i P_i \in \ker(\deg) \Leftrightarrow \sum \alpha_i = 0$

$$\bar{K}(\mathbb{P}^1) \ni \varphi = \prod_{i=1}^m (x y_i - y x_i)^{\alpha_i}$$

Пример: $x^3 + Ax + B = (x - \alpha_1)(x - \alpha_2)(x - \alpha_3)$
 $y^2 - B = (y - \beta)(y + \beta)$ $P_i = (\alpha_i, 0)$
 $\pm Q = (0, \pm\beta)$

$\text{div}(x) = [Q] + [-Q] - 2[\infty]$
 $\text{div}(y) = [P_1] + [P_2] + [P_3] - 3[\infty]$
 ↑
 Ynp! Получено

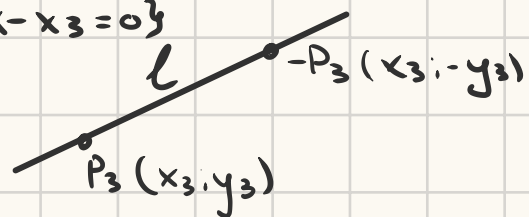
4) $\exists P_1, P_2, P_3$ на $E(\bar{k})$, $\exists$ линия на $\mathbb{P}^2$ уравн. $f(x,y) = ax+by+c$

$$\operatorname{div}(f) = [P_1] + [P_2] + [P_3] - 3[\infty] \quad \operatorname{div}(x-x_3) = \ell = \{x-x_3=0\}$$

$$\operatorname{div}\left(\frac{ax+by+c}{x-x_3}\right) =$$

$$= [P_3] + [-P_3] - 2[\infty]$$

$$= [P_1] + [P_2] - [-P_3] - [\infty] \quad \Rightarrow$$



$$[P_1 + P_2] = [-P_3] \quad \Rightarrow [P_1] + [P_2] = [P_1 + P_2] + [\infty] + \operatorname{div}\left(\frac{f(x,y)}{x-x_3}\right)$$

② Теор. $D \in \operatorname{Div}^0(E)$; тогда $D \in \operatorname{PDiv}(E) \Leftrightarrow \Phi(D) = \infty$ ($\Phi: \operatorname{Div}^0(E) \rightarrow E(\bar{k})$)

$$\text{из 4) } [P_1] + [P_2] \neq [P_1 + P_2] + [\infty] + \operatorname{div}(g)$$

$$\Phi(\operatorname{div}(g)) = \infty \quad \text{Тогда } D = \underbrace{D_{>0}}_{\text{нормальн. кр.}} + \underbrace{D_{<0}}_{\text{омпун. кр.}}$$

$$* \Rightarrow D_{>0} = [R] + t[\infty] + \operatorname{div}(g_1) \quad \text{Умнож. } \exists Q, R \in E(\bar{k}), \exists n \in \mathbb{Z}$$

$$\exists \tilde{g} \in \bar{k}(E) \text{ так, что}$$

$$\Phi(\operatorname{div}(\tilde{g})) = \infty$$

$$D = [R] - [Q] + n[\infty] + \operatorname{div}(\tilde{g})$$

$$\deg(D) = 0 \text{ (число)} = 1 - 1 + n + \underbrace{\deg(\operatorname{div}(\tilde{g}))}_0 = n = 0$$

$$\text{Тогда } D = [R] - [Q] + \operatorname{div}(\tilde{g})$$

$$\Phi(D) = R - Q + \infty = R - Q$$

$$\Leftrightarrow \exists \Phi(D) = \infty \Rightarrow R = Q \Rightarrow D = \operatorname{div}(\tilde{g}) \in \operatorname{PDiv}(E)$$

$$\Rightarrow \exists D = \operatorname{div}(f); \text{ Тогда } [R] - [Q] = \operatorname{div}\left(\frac{f}{\tilde{g}}\right)$$

$$P, Q \in E(\bar{k}) \quad \exists \operatorname{div}(h) = [P] - [Q] \text{ где нек-я } h \in \bar{k}(E)$$

$$\text{Тогда } P = Q$$

$$[P] - [\infty] \mapsto P$$

$$\text{Теорема Абеля: } \Phi \cong \operatorname{Div}^0(E) / \operatorname{PDiv}(E) \cong E(\bar{k})$$

$$\text{Пр: } E = \{y^2 = x^3 + 4x\} \text{ над } \mathbb{F}_{11}$$

$$D = [(0,0)] + [(2,4)] + [(4,5)] + [(6,3)] - 4[\infty]$$

$$\deg(D) = 0 \quad \Phi(D)$$

$$x_{P+Q} = \left(\frac{y_Q - y_P}{x_Q - x_P}\right)^2 - x_P - x_Q$$

$$y_{P+Q} = -\left(\frac{y_Q - y_P}{x_Q - x_P}\right)(x_{P+Q} - x_P) - y_P$$

$$(0,0) + (2,4) = (4-2; -4) = (2, -4)$$

$$(2, -4) + (4,5) = ((5-4)^2 - 6 = -5 = 6, 4+4) = (6, 8)$$

$$(6,8) + (6,3) = \infty \quad \underline{\Phi(D) = \infty} \quad \deg(D) = 0$$

ℓ $\ell = \{y-2x=0\}$ $(0,0) + (2,4) = (2, -4)$ $P_1 + P_2 \quad P_3 = (2,4)$
 $(0,0) \quad (2,4) - \text{кратность } 2$
 $\text{div}(y-2x) = [(0,0)] + 2[(2,4)] - 3[\infty]$

$$\begin{array}{l} (2,4) \\ (2,-4) \end{array} \left\{ \begin{array}{l} \bar{\ell} = (x-2) \\ \text{div}(\bar{\ell}) = \\ = [(2,4)] + [(2,-4)] - 2[\infty] \end{array} \right.$$

$$D = [(2, -4)] + \text{div}\left(\frac{y-2x}{x-2}\right) + [(4,5)] + [(6,3)] - 3[\infty]$$

ℓ $\frac{x-4}{6-4} = \frac{y-5}{3-5}$
 $(4,5) \quad (6,3)$
 $\frac{x-4}{2} = \frac{y-5}{-2} \quad -x+4 = y-5$
 $y = -x+9$
 $\{y+x-9=0\}$
 $\{y+x+2=0\}$

$$(x+2)^2 = x^3 + 4x;$$

$$x^3 - x^2 - 4 = 0 \quad x = 4, 6$$

$(2,4)$

$$\text{div}(x-2) = [(2,4)] + [(2,-4)] - 2[\infty]$$

$$(4,5) + (6,3) = [(2,4)] + [\infty] + \text{div}\left(\frac{y+x+2}{x-2}\right)$$

$$D = [(2,-4)] + [(2,4)] +$$

$$+ \text{div}\left(\frac{y-2x}{x-2}\right) + [\infty] - 3[\infty] +$$

$$+ \text{div}\left(\frac{y+x+2}{x-2}\right) \quad \text{div}(x-2)$$

$$D = \text{div}(x-2) + \text{div}\left(\frac{y-2x}{x-2}\right) + \text{div}\left(\frac{y+x+2}{x-2}\right) \Leftrightarrow$$

$$\Leftrightarrow \text{div}\left(\frac{(y-2x)(y+x+2)}{x-2}\right) = \text{div}\left(\frac{x^2+2y-xy-2x^2}{x-2}\right) = \text{div}\left(\frac{(x-2)(x^2-y)}{x-2}\right) = \text{div}(x^2-y)$$

Спаривание Ве́йна

$$\Phi: \mathcal{C}^0(E) \xrightarrow{\cong} E(\bar{K})$$

$$\Phi: \mathcal{C}^0(E)[m] \xrightarrow{\cong} E(\bar{K})[m] \quad \leftarrow \text{точка } m\text{-кривой} \quad (mP=0)$$

↓
 гомоморфизм, то
 отображение на m -кривую.

$$f \in \bar{K}(E), D = \sum a_i [P_i]$$

$$f(D) = \prod_i f(P_i)^{a_i} \in \bar{K}$$

$$\text{Согласно, если } \text{div}(f) \cap \text{Supp}(D) = \emptyset$$

$$\{P_i \mid a_i \neq 0\}$$

$$\Phi_{\text{Poin}} \left(\begin{smallmatrix} \text{Bzaimnosc} \\ \text{Beine} \end{smallmatrix} \right) f_1, f_2 \in \bar{K}(E)$$

$$f_1(\text{div}(f_2)) = f_2(\text{div}(f_1))$$

$$D_1, D_2 \in \text{Div}^0(E)[m], \text{ так что } m \Phi(D_i) = \infty$$

$$\Phi''(mD_i)$$

$$\text{По Th Asera } \exists f_1, f_2 - \text{pay } \in \bar{K}(E) : mD_1 = \text{div}(f_1) \text{ и } mD_2 = \text{div}(f_2)$$

$$\exists \text{Supp}(D_1) \cap \text{Supp}(D_2) \neq \emptyset$$

$$\langle D_1, D_2 \rangle = \frac{f_1(D_2)}{f_2(D_1)} \in \bar{K}$$

$$\text{Сб-ба: } \langle D_1, D_2 \rangle^m = \frac{f_1(D_2)^m}{f_2(D_1)^m} = \frac{f_1(mD_2)}{f_2(mD_1)} = \frac{f_1(\text{div}(f_2))}{f_2(\text{div}(f_1))} \xrightarrow{\text{Beine}} 1, \text{ т.е.}$$

$$\text{Im } \langle, \rangle \subset \mu_m(\bar{K}) = \{x \in \bar{K} \mid x^m = 1\} \text{ группа корней m-й ст.}$$

$$\langle, \rangle - \text{кор-но опрег на } \mathcal{C}^0(E)[m]$$

$$3) \langle D_1, D_2 \rangle = \frac{f_1(D_2)}{f_2(D_1)} \quad \langle D_2, D_1 \rangle = \frac{f_2(D_1)}{f_1(D_2)} \quad \langle D_1, D_2 \rangle = \langle D_2, D_1 \rangle^{-1}$$

$$4) \text{ (Бундесинг)} \exists mD_1 = \text{div}(f_1) \quad \langle D_1 + D_1', D_2 \rangle \equiv$$

$$mD_1' = \text{div}(f_1') \quad mD_1 + mD_1' = \text{div}(f_1 \cdot f_1')$$

$$mD_2 = \text{div}(f_2)$$

$$\equiv \frac{(f_1 \cdot f_1') D_2}{f_2(D_1 + D_1')} = \frac{f_1(D_2) \cdot f_1'(D_2)}{f_2(D_1) f_2(D_1')} = \langle D_1, D_2 \rangle \langle D_1', D_2 \rangle$$

$$\langle D_1, D_2 \rangle \langle D_1', D_2 \rangle$$

$$\exists m \nmid 2 \langle P, P \rangle \mid \langle P, P \rangle^m = 1. \quad \langle P, P \rangle^{-1} = \langle P, P \rangle = \langle P, P \rangle^2 \Rightarrow \langle P, P \rangle = 1$$

P-точка

5) Небундесингенов:

$$\text{Если } P = \infty, Q \notin \langle P \rangle, \text{ то } \langle P, Q \rangle \neq 1$$

$$\mathcal{C}^0(E)[m] \times \mathcal{C}^0(E)[m] \xrightarrow[\#]{\langle, \rangle} \mu_m(\bar{K})$$

$e_m: E(\bar{K})[m] \times E(\bar{K})[m] \rightarrow \mu_m(K)$ — сопоставление Вейля,
индуциров. $\langle, \rangle$ с нормой из-а $\mathbb{Q}$

Как вычислить?

$$P, Q \in E(\bar{K})[m]: D_P = [P] - [\infty] \leadsto f_P \in \bar{K}(E)$$

$$D_Q = [Q] - [\infty] \leadsto f_Q \in \bar{K}(E)$$

Заменяем D_Q на $\tilde{D}_Q = [Q+S] - [S], S \in E(\bar{K}) \setminus \{\infty, P, -Q, P-Q\}$
 $\mathbb{Q}(\tilde{D}_Q) = \mathbb{Q}; \tilde{f}_Q \in \bar{K}(E)$

$$e_m(P, Q) = \langle D_P, \tilde{D}_Q \rangle = \frac{f_P(\tilde{D}_Q)}{\tilde{f}_Q(D_P)} = \frac{\frac{f_P(Q+S)}{f_P(S)}}{\frac{\tilde{f}_Q(P)}{\tilde{f}_Q(\infty)}} = \frac{\left(\frac{f_P(Q+S)}{f_P(S)} \right)}{\left(\frac{\tilde{f}_Q(P-S)}{\tilde{f}_Q(-S)} \right)}$$